Learning Objective 6.7: I can receive and decode ADS-B messages from aircraft transponders.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate).

1. **Reading the chips.** A receiver samples a message at 2 MS/s, so each 0.5 µs half of a bit is one sample, or chip. After the preamble, the first 16 chips it reads, in pairs, are

   `10 01 01 01 10 10 01 10`

   (a) Write the 8 bits these chips carry, and the byte in hex.

   bits: ____

   hex: ____

   (b) The first 5 bits are the Downlink Format and the next 3 the capability. Find both, and say whether this is an ADS-B extended squitter.

   DF: ____

   CA: ____

   (c) A noise spike makes the sixth pair read `11`: both chips high. In words, explain what the receiver does with that bit, and why the parity check matters here.

2. **The message.** The Mode S Decoder prints

```
8DAE07C220483234D71CA031ACD8
```

The 28 hex digits are the 112 bits: DF and CA (8 bits), ICAO (24), ME (56), and PI (24).

(a) Split the message into its ICAO address, ME field, and parity, in hex.

ICAO:

PI:

(b) The type code is the first 5 bits of ME. Find it, and say which kind of message this is.

(c) After the type code and 3 category bits, the callsign is eight 6-bit characters. In the Mode S character set, 1 to 26 are A to Z, 48 to 57 are the digits 0 to 9, and 32 is a space. The eight values are 18, 3, 8, 52, 53, 49, 50, 32. Write the callsign.

3. **Receiver settings.** The L18 receiver runs at 2 MS/s.

   (a) How many samples does one whole message take, preamble included? Set the triggered time sink's number of points to show one whole message with about 10 µs to spare.

   samples: ____

   points: ____

   (b) A classmate sets the RTL-SDR to 1 MS/s to save processing. Explain, in words, why the Mode S Decoder can no longer read the bits.

   (c) On a live run, the 20 ms plot shows a noise floor near 0.05 and messages whose pulses reach between 0.15 and 0.6. The trigger level is 0.3. Choose a better trigger level and say what you would see with each setting.

4. **What noise does.** You run `L18-adsb-rx.grc` on the simulated recording and raise the Added noise slider a step at a time.

   (a) Describe what happens to the 20 ms plot, the Messages label, and the aircraft table as the noise rises.

   (b) Explain, in words, why the weakest aircraft drop out first, and what that means for a real receiver.

5. **Generating and locating.** Anyone with an RTL-SDR can receive ADS-B, and a flowgraph can generate it.

   (a) The L18 generator writes a file. Explain why nothing in the course transmits its output at 1090 MHz.

   (b) The Mode S Decoder finds each position from one message by using USAFA as a reference position. Explain, in words, why it needs a reference at all, and what goes wrong for an aircraft 500 nautical miles away.